

CMSC 478: Assignment 1

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Manual Calculation for the Informational Gain

Entropy: $Entropy(S) = -\sum_{i=1}^n P_i \log_2 P_i$

There are Yes=8, No=8, Total=16

Proportion of positive example: $8/16 = 1/2$

Proportion of negative example: $8/16 = 1/2$

$Entropy(S) = H(S) = -(p \log_2 p + (1-p) \log_2 (1-p))$
 $= -(\frac{1}{2} \log_2 (\frac{1}{2}) + (1-\frac{1}{2}) \log_2 (1-\frac{1}{2}))$
 $= 1$

Informational Gain

Remote (Yes Group): Yes=4, No=4, Total=8

p positive: $4/8 = 1/2$; p negative: $4/8 = 1/2$

$H(Yes) = -(p \log_2 p + (1-p) \log_2 (1-p))$
 $= -(\frac{1}{2} \log_2 (\frac{1}{2}) + (1-\frac{1}{2}) \log_2 (1-\frac{1}{2})) = 1$

Remote (No group): Yes=3, No=5, Total=8

p positive: $3/8$; p negative: $5/8$

$H(No) = -(\frac{3}{8} \log_2 (\frac{3}{8}) - (\frac{5}{8}) \log_2 (\frac{5}{8})) = .954434002921$

$Gain(S, Remote) = Entropy(S) - ((S_{Yes}/|S|)H(Yes) - (S_{No}/|S|)H(No))$
 $= 1 - (8/16)(1) - (8/16)(.954434002921)$
 $= .0227829986 = .023$

Salary (Yes Group): Yes=4, No=4, Total=8

$H(Yes) = 1$

Salary (No group): Yes=3, No=5, Total=8

$Gain(S, Salary) = .023$

Tech Stack (Modern Group): Yes=7, No=1, Total=8

p positive: $7/8$; p negative: $1/8$

$H(Modern) = -(\frac{7}{8} \log_2 (\frac{7}{8}) - (\frac{1}{8}) \log_2 (\frac{1}{8})) = .5435644432$

Tech Stack (Legacy Group): Yes=0, No=8, Total=8

p positive = 0; p negative: 1

$H(Legacy) = -(0) \log_2 (0) - (1) \log_2 (1) = 0$

$Gain(S, TechStack) = 1 - (8/16)(.5435644432) - (8/16)(0)$
 $= .7282177784$

Questions from Chapter 1 on Decision Trees in CIML Book (25 points)

Q1: Critique the greedy decision-tree feature scoring rule: The training procedure selects the next feature using a majority-vote accuracy score on the NO/YES splits (a greedy approach). Explain why this greedy criterion can be appealing, then argue when it might lead to a worse final tree than another feature choice. Your answer should include a plausible dataset situation (you can invent it) where a locally “best” split harms global performance.

It can be appealing because it can be applied to many learning problems and it is simple. It evaluates one split at a time. The Greedy decision-tree might lead to a worse final tree than another feature because it chooses a feature with the best immediate majority vote. It does not consider the future effects of the split. One feature might have a low immediate score but later in the tree have a better score. Since it has a low immediate score, the greedy scoring rule ignores it.

Q2: Distribution shift and “learning the wrong lesson.” Discuss the tanks example as a cautionary tale: the classifier succeeded on held-out data but failed in the real world due to a mismatch in what it actually learned. Explain what kind of bias this illustrates, and propose two concrete safeguards you would implement in dataset collection or evaluation to reduce the chance of this failure mode.

The US Government managed to build a classifier with nearly 100 percent accuracy using held-out data but failed to run in the real world. It was unable to distinguish between US and Russian tanks but just between clear and blurry photos. According to The Neural Net Tank Urban Legend (<https://gwern.net/tank>), the story warns about dataset bias. I would implement a diverse collection of datasets and train the model with these datasets to reduce the chance of this failure mode.

Q3: How does ID3 use information theory to improve tree building?

ID3 uses entropy and information gain from information theory which improves tree building. ID3 algorithm constructs decision trees by selecting, at each node, the attribute that maximizes information gain. Entropy measures that uncertainty in the dataset, and information gain measures how much that uncertainty is reduced after a split. ID3 leads to purer splits and more structured trees by choosing the feature that most reduces entropy.

Q4: Why train/test distributions must match (failure case analysis). Recount the “US tanks vs. Russian tanks” cautionary example and explain what went wrong in terms of distribution mismatch and spurious cues. Then propose two concrete fixes to the data collection or evaluation process that would better ensure the learned model captures the intended concept.

The US tanks vs. Russian tanks cautionary example tells a story of how misleading features and distribution mismatch can lead a classifier to fail in the

real world. To ensure the learned model captures the indented concept, I would collect diverse data set and evaluate the model by testing it with the diverse data set.

Q5: Memorization vs. generalization (conceptual + example). Explain the difference between memorization and generalization in supervised learning. Use the course recommendation example to describe (i) an unfair test scenario, (ii) a too-easy test scenario, and (iii) what a reasonable test should look like.

The difference between Memorization and Generalization is that Memorization is what you know and generalization is based on taking what you know and apply it to new problems. An example of memorization would be (ii) scenario and an example of generalization is scenario (iii) (i) An unfair test scenario would be asking questions on something that was not talked about the during the entire semester. The book talks about Bob giving Alice an exam on History of Pottery when he spent an entire semester talking about machine learning. (ii) A too-easy test scenario would be Bob asking Alice exactly the same questions that he has answered during lectures. (iii) A reasonable test would be testing Alice with new problems based on the lectures she observed.